

The following questions can help you think critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

First, I examined what the algorithm had to do with the three-digit numbers. The task was to extract each digit from the three-digit number. This means that there would be a need for the calculation of hundreds, tens, and ones digits separately.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Initially, it was hard for me to figure out how division and the % operator would divide the digits. To understand how the calculations were done, I experimented with various three-digit numbers.

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

I decomposed the problem before coding. I planned to find the hundreds place digit first, followed by the tens place digit, and the ones place digit. Finally, all the digits would be displayed. What also helped was the fact that I had done a previous skillbuilder that displayed both the tens and ones place. I ended up recycling in this mastery code to help.

How did you decide on the tools, data structures, or algorithms to use?

“Scanner” is used since I had to retrieve an integer input from the user. “int” variables are used since both the number and the digits were whole numbers. Division and % are used since they were helpful in retrieving each digit.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

The code was written piece by piece instead of all together. First, I wrote the code for the Scanner and the user input for the number, followed by the calculation for each digit, and finally I added the code for displaying the output.

How did you test your solution along the way to make sure it was working?

I ran this program using various three-digit numbers. I compared the hundreds, tens, and units digits with those that I entered into the program to ensure their accuracy. I also tested each code individually to see if it worked correctly.

Overcoming Challenges

What part of the problem was the most difficult for you?

The hardest part for me was understanding how to find the tens number. Finding the hundreds and ones numbers seemed easier than finding the tens number since I needed to use both % and division. After trying different examples, I got it right.

How did you handle moments when you felt stuck or unsure of what to do next?

Whenever I found myself stuck, I divided the problem into smaller calculations. I then tested each calculation using a number and determined the output value for the calculation. It helped me understand the program rather than just blindly guessing it. I also used AI to help give me ideas on what the code should end up like so that it was accurate.

Learning

Was there anything you learned that you think will help you with future challenges?

I found out that using division and the % operator could help extract digits from a number. In addition, I discovered that by dividing a program into smaller pieces, we make the solution to the program much easier. I believe these skills will help me with future Java programs.